

Neutron star - White Dwarf (NSWD) binaries formation and evolution as a multimessenger source.

Master's Thesis

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Abstract

Neutron Star - White Dwarf Binaries (NSWD) are systems which are of great importance in the newly developing field of Multi-Messenger Astronomy and understanding their formation and evolutionary pathways can provide key insights to enhance our understanding of the universe. In our study we explore formation channels of these systems and their evolution channels leading to their mergers and post-merger nucleosynthesis modelling. Future of multi-messenger astronomy is bright in all the different spectrums as the new electromagnetic (em) observatories are going live with the future space-based Gravitational wave (GW) observatories like LISA, DECIGO, B-DECIGO in their development stage and neutrino detection facilities like Icebox already in their observation runs with improvements to it already under development. These will open us up to a new era where systems that were hidden from us will be observable, one of such systems will be, NS-WD systems which will produce GW in mhz - decihz frequency range close to their mergers which will be observable by space based GW detectors like LISA and DECIGO. We study population synthesis of such systems using binary stellar evolution code `binary_c`, where we statistically model evolution pipeline and study the rate of formation of such systems evolving them from Zero age main sequence (ZAMS) to see which combination of masses and periods form them.

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1 Introduction

Gravitational waves (GWs) have opened a new window onto the universe, allowing us to observe cosmic phenomena through the ripples in spacetime they produce. Predicted by Einstein in 1916, GWs were first directly detected in September 2015 by the Advanced LIGO detectors. This landmark detection, arising from a binary black hole merger, confirmed a century-old prediction and marked the birth of gravitational wave astronomy. Since then, dozens of compact object mergers (black holes and neutron stars) have been observed via their GW “chirp” signals – waveforms that increase in frequency and amplitude as the two objects [spiral together](#). These observations not only validate general relativity but also provide unprecedented insights into the properties of black holes and neutron stars.

While the current observations have been limited by the sensitivity of LIGO-VIRGO-KAGRA (LVK) collaboration, which measure GW in the frequency range 20 Hz to 20,000 Hz with sensitivity reaching to 10 Hz in the upcoming years(1), the future of GW astronomy will include space-based GW detectors like LISA which will be sensitive in the mhz band.(2) These detectors will open up window to an entire new ways to study compact objects and specifically types of binaries which are not visible in the sensitivity range of LIGO - Binary white dwarfs binaries (**BWDs**), Neutron star - White Dwarfs binaries(**NS-WD**. NS-WD binaries consist of a neutron star (an ultra dense stellar remnant) and a white dwarf (a compact remnant of a lower mass star) in orbit around each other. If the orbital separation is small enough, this system will produce gravitational radiation that cause the orbital radius to gradually shrink, overtime this can lead to WD to spiral inward toward the NS. The GW from such an inspiraling NS-WD would be in the low frequency range (mhz to decihz), making the space based observatories like LISA, DECIGO etc. Recent population studies by Toonen et al(3) predict that LISA could detect on the order of 10^2 NS-WD binaries in our galaxy, more optimistically decihz observatories may detect thousands of NS-WD merger events per year. Although no NS-WD inspiral has been yet detected via GWs, yet there is indirect evidence that these systems emit GW. For example, the pulsar-white dwarf binary **PSR J0348+0432** has an orbital period of only 2.46 hours, and radio timing measurements have revealed that its orbit is decaying at precisely the rate predicted by General Relativity due to GW emission. (4). This mirrors the earlier discovery in the Hulse-Taylor binary pulsar (two neutron stars orbiting each

other) where the measured orbital decay (about 76 microseconds per year) matched the GW radiation theory to within 0.2% which earned them the 1993 Nobel Prize in Physics. (5)

NS-WD mergers are not only important as GW sources, but they may also produce unique electromagnetic (em) signatures, making them a highly interesting target for multi-messenger astronomy. If a WD is disrupted or accreted by a neutron star, which as a process is very likely following their evolution and inspiral, it could trigger explosive transients like Gamma ray bursts (GRBs) or Type I supernovae (SNe), the mechanism of which is still a mystery and a topic of study. In one such peculiar long Gamma ray burst, (GRB211211A), which was followed by a faint "Kilonova-like" optical transient, has been suggested as a possible NS-WD merger event(6) . If this interpretation is correct, it implies that NS-WD mergers can produce both gravitational waves and electromagnetic flashes. Future GW detections with early warning (as decihertz detectors could provide) would enable telescopes to prepare and catch such counterparts. Clearly, studying NS-WD binaries is crucial for understanding their role in the cosmos – from their contribution to the low-frequency GW background up to their potential as progenitors of exotic transients.

However, to fully leverage the potential of these cosmic laboratories (NS-WD binaries) we must theoretically predict their gravitational wave signals, their **formation, evolution and merger rates**. Unlike Binary Black holes (BBHs) and Binary Neutron Star (BNS) mergers, NS-WD systems involves a WD whose structure, mass and Radius can vary widely (7) and complex interactions of NS with WD prior to their mergers would also affect their inspiral phase, including the merger phase. Most of the NS-WD binaries are a product of binary stellar evolution -> e.g. a massive star goes supernova to form a NS while the companion star evolves into a WD). The evolutionary path often involves mass transfer or common-envelope phase that brings the two remnants closer together. The diversity of outcomes - in terms of masses, orbital periods, eventual merger vs long-term stability- means we need detailed modeling to predict the rate at which these mergers happen and the associated GW they will emit.

2 Literature Review

To understand and create physically accurate populations using any technique, we must understand the statistics of the population itself, which implies knowing given a set of initial conditions like the beginning mass of a cluster or the universe what would be the current population of NS-WD binaries in our galaxy as well as the universe. As lighter mass binaries such as BWD would be much more abundant than extreme binaries like BBH, BNS and NSBH and ofcourse, NS-WD.

2.1 Initial Mass Function (IMF) and IMF universality theorem

The initial mass function (IMF) describes this statistical distribution of masses. In the canonical formulation done by Kroupa et al 2008(8), the IMF is a broken power law with a shallow slope ($\alpha_1 \approx 1.3$) between 0.08 and 0.5 $M_\odot$ and a steeper slope ($\alpha_2 \approx 2.3$) between 0.5 and 150 $M_\odot$ (8). These slopes define a universal IMF (the IMF universality Hypothesis) that, after corrections for unresolved binaries, Brown dwarfs (BWd), appears to hold across a wide range of star-forming environments. The Cluster IMF theorem states that IMF cannot be definately extracted for any star cluster (8). It follows that all known well-resolved stellar populations are consistant with having been drawn from one parent distribution, the IMF universality Hypothesis therefore cannot be discarded, despite the Cluster IMF Theorem. This means that the current star-forming theory fails to predict stellar outcome because it depends on the IMF being dependent on physical boundary conditions not currently observed(8). The Intergalactic initial mass function, IGMIF Theorem, however predicts a variation of galaxy-wide IMFs due to galaxy's star formation rate even if the IMF univerality hypothesis is valid. The variation has been confirmed by Sloan Digital Sky Survey (SDSS) galaxy data (8)

2.2 Binary population study for formation and evolution of Neutron Star - White Dwarf (NS-WD) binaries

Toonen et al. (2018) (3) performed a comprehensive binary-population study (BPS) of NS-WD binaries. Their models explored different assumptions about common-envelope (CE) evolution and natal kicks. They found most NS-WD mergers arise where WD forms before NS and where mass transfer plays a critical role. Carbon-Oxygen (CO) WD

comprise of 60-80% of the merging WDs, with the remainder being Oxygen-Neon WDs. The mass transfer often involves a first phase that is stable followed by CE phase. For a canonical system with a $1.3 M_{\odot}$ neutron star (NS) and a $0.8 M_{\odot}$ white dwarf (WD) in a circular $1 R_{\odot}$ orbit, gravitational-wave emission causes the two objects to merge in ~ 70 Myr; if the orbit has an eccentricity of 0.5 (0.75), the inspiral time drops to 25 (14) Myr. When the WD fills its Roche lobe, mass transfer can be either stable (leading to an ultra-compact X-ray binary) or unstable. Assuming a critical WD mass of $0.2 M_{\odot}$, over 99% of semi-detached NS–WD systems merge. Natal kicks rise to wide spatial offsets: some mergers occur hundreds of parsec away from the host galaxy. The mergers resemble peculiar type Ic-like supernovae, predominantly in late-type galaxies.

2.3 GW observations from NS-WD mergers

Yin et al (9) explored how GW observations can distinguish between merger types by examining the frequency domain. They notes that NSBH mergers emit at almost 100 Hz, While NS-WD mergers produce signals near 0.1 Hz, which is the decihz band, these systems will be detectable by future space based GW detectors like DECIGO and the proposed decihz detector by India. Decihertz detections would provide early warnings and enable multimessenger follow-up. The frequency difference thus offers a way to identify NS–WD mergers purely from their GW signal, complementing electromagnetic classification.

2.4 Merger and Post merger nucleosynthesis

During merger and post merger analysis, the composition of merger ejecta determines whether a NS-WD merger will produce a kilonova or other transient. Katenborn et al. (2022) (10) used Popsynth and hydrodynamic modelling to follow NS-WD mergers in which the WD undergoes unstable mass transfer and complete tidal disruption. It forms a large accretion disk around the NS, nuclear burning, neutrino emission and disk winds were followed self-consistently. They found that outflows are rich in iron and nickel, powering bright optical and infrared transients (10). These “super-nova-like” events are short-lived but may reach high luminosities and could be accompanied by gamma-ray bursts Compared to BH–WD mergers, the mass ratio in NS–WD systems produces disks with larger radii and slower accretion rates, leading to unique transient signatures (10).

Although NS-WD mergers produce significant amounts of material, they may contribute little to the cosmic inventory of heavy r -process elements. Chen et al. (2024) (11) estimated total r -process yields from various merger types suggest that binary neutron-star mergers alone can explain the Milky Way’s heavy r -process content, whereas NS-WD mergers are unlikely to eject enough neutron-rich matter. It is argued that NS-WD mergers produce $\lesssim 0.005 M_{\odot}$ of r -process material, far below the amount needed to influence galactic chemical evolution. This implies that, while NS-WD mergers may show nucleosynthetic signatures (e.g., iron-group elements), they are not a major site of heavy r -process nucleosynthesis.

2.5 Observed NS-WD system

Observationally, NS-WD systems are only occasionally confirmed, as it would be very hard to distinguish a single WD from a binary WD paired with NS, as the size and luminosity of WD will overshadow its companion and therefore, NS-WD have rarely been observed or confirmed.

Bhalerao et al. (2017) (12) reported the optical discovery of the $0.5 M_{\odot}$ white-dwarf companion to the $2 M_{\odot}$ millisecond pulsar PSR J1614–2230, which is a mystery in itself as to why the mass of the NS in this system is this high, as most NS observed typically have a mass of $1.2 - 1.4 M_{\odot}$. Photometry showed that the companion is a He-CO white dwarf aged 2.2 Gyr. The inferred accretion rate during the spin-up phase was less than 10^{-2} of the Eddington rate, and the pulsar’s spin period was close to its birth value. These properties challenge simple recycling models of millisecond-pulsar formation and demonstrate how precise mass and age measurements of NS-WD systems can constrain accretion histories and evolutionary paths. Observations like PSR J1614–2230 also provide calibration points for population-synthesis predictions.

2.6 Modelling Binary Evolution: Rapid Algorithms and Detailed Codes

Comprehensive modelling of binary evolution is essential to predict NS-WD populations and their observable signatures. Hurley, Tout & Pols (2000) (13) developed a rapid binary evolution algorithm that incorporates mass transfer, accretion, common-envelope

evolution, supernova kicks, angular-momentum loss, and tidal interactions. The authors showed that including tidal circularisation and synchronisation can significantly alter population outcomes; ignoring tides leads to systematic errors in predicted binary fractions. They recommended sampling initial separations using the semi-latus rectum distribution rather than orbital periods to better match observations. Rapid algorithms like this underlie many BPS studies, but their approximations (e.g., simplified stellar models) limit their accuracy.

2.7 Using MESA for population synthesis

While rapid algorithms offer statistical efficiency, detailed one-dimensional codes like (Modules for Experiments in Stellar Astrophysics), MESA evolve stars and binaries with more physics (e.g., adaptive mesh, nuclear networks, time-dependent convection). MESA allows one component of a binary to be treated as a point mass—ideal for modelling NS–WD systems—while evolving the companion’s structure under mass transfer and gravitational-wave angular-momentum loss. In the context of NS–WD research, MESA can trace the evolutionary history of the progenitor stars, the onset of mass transfer, and the growth of the WD’s radius up to Roche-lobe overflow. Combining MESA’s detailed evolution with population-synthesis models like `binary_c` will allow us to create a better population of NS-WD binaries, enabling more accurate predictions of NS–WD merger rates gravitational-wave signals and transient properties. Toonen et al (2008) (3) have found that mass transfer plays a critical role in the evolution of NS-WD systems and in most of the confirmed formation of a NS-WD system the WD forms first.

3 Modeling a Population of NS-WD Binaries

3.1 Sampling the IMF to estimate Compact-Binary Fractions

To explore how often compact-binary pairs form under the canonical IMF, we implemented a Monte-Carlo code that samples primary masses (M_1) from the Kroupa IMF (8) and secondary masses ($M_2 = q, M_1$) from a uniform mass-ratio distribution $0.01 \leq q \leq 1$. A toy remnant-mapping function was applied: stars below $8 M_\odot$ become white dwarfs (WDs), stars between 8 and $15 M_\odot$ become neutron stars (NSs), and more massive stars collapse to black holes (BHs). Sampling 100 million binaries produced the following approximate mixture: double-WD systems (BWD) dominate at $\approx 99\%$; NS–WD systems (NSWD) constitute $\sim 0.27\%$; BH–WD $\approx 0.084\%$; double-NS (BNS) $\approx 0.08\%$; NS–BH (NSBH) $\approx 0.07\%$; and double-BH (BBH) $\approx 0.1\%$. In other words, NS–WD pairs are the most common double-compact binaries after WD–WD pairs, and they are more abundant than NS–BH or double-NS binaries. Although this toy model neglects binary interactions, it illustrates how the IMF strongly favours low-mass primaries and thus biases populations toward WD-containing systems.

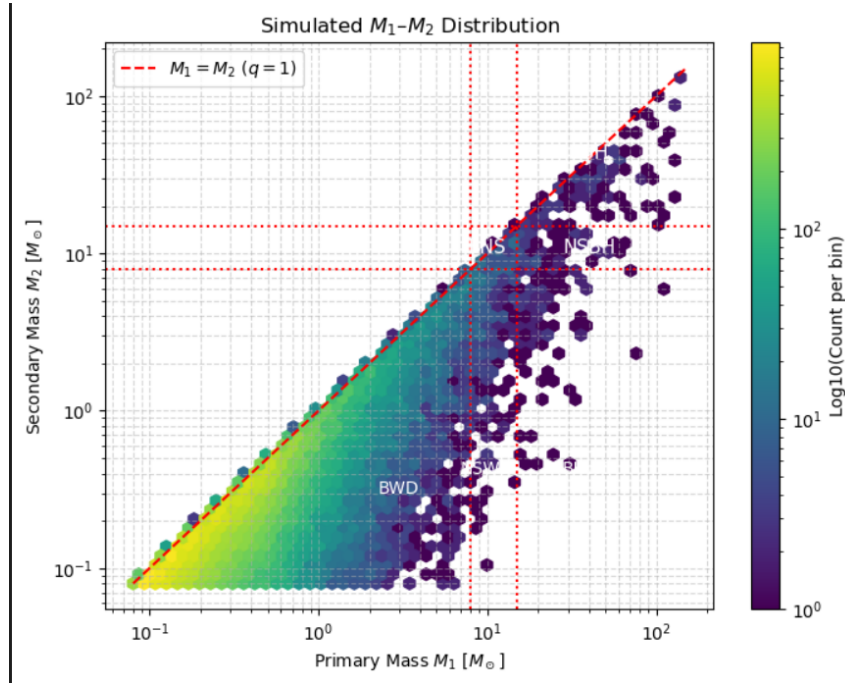


Figure 1: Fraction of different types of compact binary systems sampled from population of 100 million using IMF from kroupa et al. (8).

3.2 Binary Population synthesis

Traditionally to generate a population of compact binaries involving systems like BBH, BNS etc. a simple approach is used, which is to generate millions of binaries with random initial conditions and evolve them with simple physics, yielding statistical predictions for merger rates and distribution of the population. While this is efficient, this method uses approximate prescriptions complex processes like mass transfer, common-envelope evolution, winds and tidal interactions. In contrast a detailed simulation approach can track the evolution of a binary with higher physical fidelity. In this thesis, we adopt a detailed modeling strategy centered on the **Modules For Experiments In Stellar Astrophysics MESA**) code to create and study a synthetic population of NS-WD binaries.

3.3 Modules For Experiments In Stellar Astrophysics (MESA)

[MESA](#) is a state-of-the-art open-source Stellar evolution code that has become a cornerstone of computational Astrophysics. It is one-dimensional (1D) stellar evolution toolkit capable of modelling a wide range of scenarios from pre-main-sequence stars to compact objects. For our purpose, the relevant part of the code is the binary evolution module (MESAbinary), which allows two stars (or a star + compact object) to be evolved together with processes such as Roche-lobe overflow, mass transfer, tidal disruptions, exchange of orbital angular momentum, angular momentum losses etc. MESA's capabilities include detailed treatment of stellar interiors (with adaptive mesh refinement and modern input physics) and the ability to handle events like Roche-lobe overflow and accretion onto a neutron star. For our purpose, we could treat the NS or even the WD as a point mass and MESA will self-consistently account for mass transfer when a star fills the other's Roche lobe and how the orbit would change in response.

A key aspect of evolution using MESA is that it includes loss of orbital angular momentum via GW radiation. Using the general relativistic formula for GW radiation. (14) MESA can calculate the rate at which the orbit shrink due to GW emission. Physically, as the NS-WD binary orbits, it emits GW and loses energy and angular momentum, causing the separation and orbital period to decrease over time. MESA's binary module applies this effect continuously during the evolution, thus even in the absence of mass transfer, a NS-WD binary in MESA will gradually inspiral on a timescale that depends on its masses and separation. By including both GW losses and any mass transfer inter-

actions, MESA can follow an NS-WD system from wide separations down to very close orbits. If the white dwarf begins to transfer mass to the neutron star, MESA will also track the mass loss from the WD and the response of its radius, which is critical for determining whether the mass transfer is stable or leads to a runaway merger.

3.4 `binary_c`

`binary_c` software framework models the evolution of single, binary and multiple stars, including stellar evolution and nucleosynthesis. It is designed for stellar population calculations, so is lightweight and versatile, and its support software contains tools for development and data analysis. `binary_c` is happy in standalone, virtual and HPC environments. Exploration of this code is not yet done up until this point in this project, next goal will be to integrate the population generated by this method and by using MESA, to get a comparison between ratio of population for different systems and the timeline for their evolutions.

3.5 Using MESA to generate NS-WD systems

3.5.1 Setting up MESA in system and Cluster (PARAM_RUDRA)

Using MESA in a personal machine or a cluster requires intricate setup of MESA itself in the system, in cluster it is different as we don't have admin access to directly install required libraries that MESA uses so we use other methods or try to work-around these difficulties using different methods. We start by downloading the required libraries by installing `mesasdk` which contains all the libraries used by MESA in our system by directly downloading the .tar file and unzipping it. After it is downloaded we install it in our system using `.install` and install it in the specified path. After `mesasdk` is installed in system we directly download MESA either using `gitclone` or `wget` from the official website and put it in a specific path. After the download we install MESA in our system. After both MESA and `mesasdk` have been installed we setup the source file (.sh) to source MESA in our system using a specific source file name like `profile_mesa.sh`, this is done to make sure MESA environment is not active on the default login, that is why don't put it in the `.bashrc`, to avoid conflicts with other programmes. In gist this is the setup we need to do, doing this required me to learn terminal and how it worked with the commands as

this was my first time working in a terminal based environment.

Setting up MESA on IIT Bombay cluster, PARAM_RUDRA required a similar structured steps were performed with the included burden of setting up libraries by transferring them from personal system to cluster using -scp command. A key failure to running MESA in cluster was the use of gitclone to download the latest unreleased version of MESA, this was giving unexpected errors that took decent time to fix.

4 Results

4.0.1 First run on system

The first run was done using personal machine which has an intel i7 9th gen CPU with 6 cores, it was started on 3rd of october 2025 and ended after 7-9 hours of running. The initial masses of the two stars to be evolved were $M1=1 M_{\odot}$ and $M2 = 3 M_{\odot}$ with initial period of 5 days, the system was evolved keeping 1 star as a point and evolving the other star self-consistently. This system evolved for 100 billion years. The plots were prepared by analysing the data in binary_history.data using custom python pipelines that have ability to create plots for a single binary system and the profile of the core from the history.data file of the star we chose to evolve. Multiple plots for multiple systems can also be created on the same plot using custom-created pipeline which compares evolution of many different systems and plots them in a single plot. This was done later in the project when runs were started to be done on cluster.

The inlist_project file of this runs looks like below

Table 1: Binary Star Simulation Parameters

Parameter	Value	Description
inlist_names(1)	'inlist1'	Inlist for primary star
inlist_names(2)	'inlist2'	Inlist for secondary star
evolve_both_stars	.false.	Evolve only one star
m1	1.000000d0	Donor mass (in $M_{\odot}$)
m2	3.000000d0	Companion mass (in $M_{\odot}$)
initial_period_in_days	5.000000d0	Initial orbital period
limit_retention_by_mdot_edd	.true.	Limit accretion by Eddington rate
max_tries_to_achieve	20	Max iterations to achieve convergence

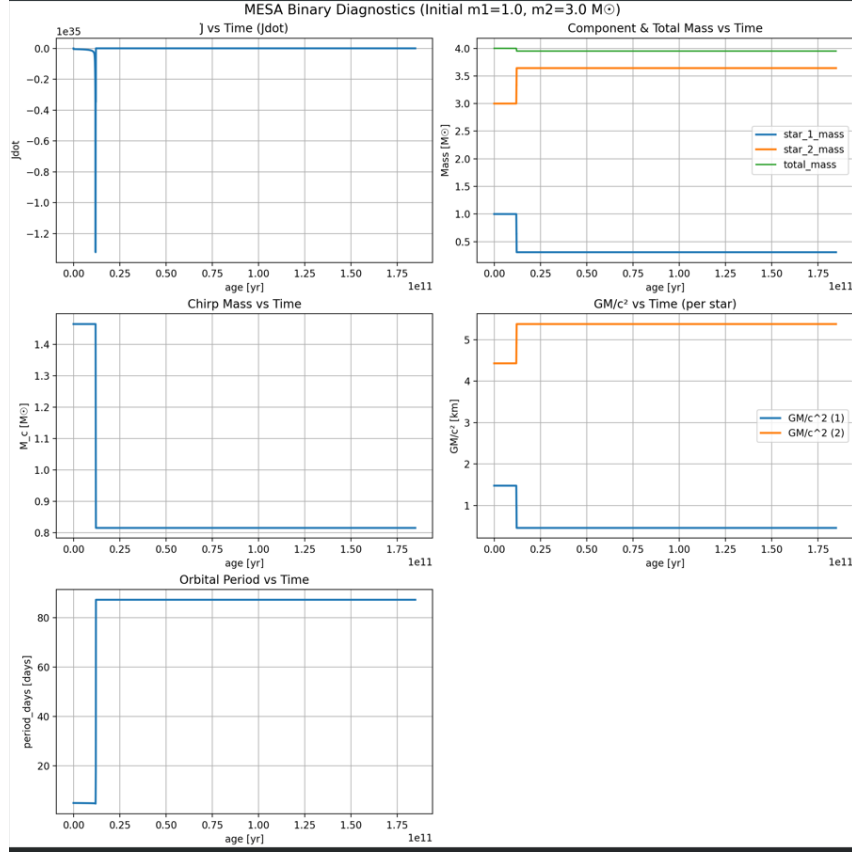


Figure 2: Parameter evolution of a binary system with $M_1=1 M_\odot$ and $M_2=2 M_\odot$ using MESA.

We have put the `evolve_both_stars` flag to be `.false`. this means only one of the star will evolve physically changing it's inside layers as Hydrogen fuses and forms helium followed by Helium (He) flash which will turn the entire core into a He core, and hence a white dwarf. Only the $3 M_\odot$ star is massive enough to cause this and hence it is evolved, the other star is treated as a point the system is evolved as binary system. Observing the masses of each star we can see there is an initial mass transfer between the stars probably due to some wind but after that the system becomes stable and no further mass transfer is observed for it, given the small initial masses this makes sense, since one star will form a WD earlier and other one will form a WD later but given the age of simulation we see no further change in the total mass or mass transfer between stars. This is backed by comparing the mass vs age plot with period vs age plot, the period stays constant for a while with a sharp decrease for sometime, after which it increases very sharply and becomes constant throughout the evolution of the system. The constant period indicates the system is now in a stable equilibrium state with no transfer of angular momentum j_dot , decrease in period and no mass transfer - hence, a stable binary system.

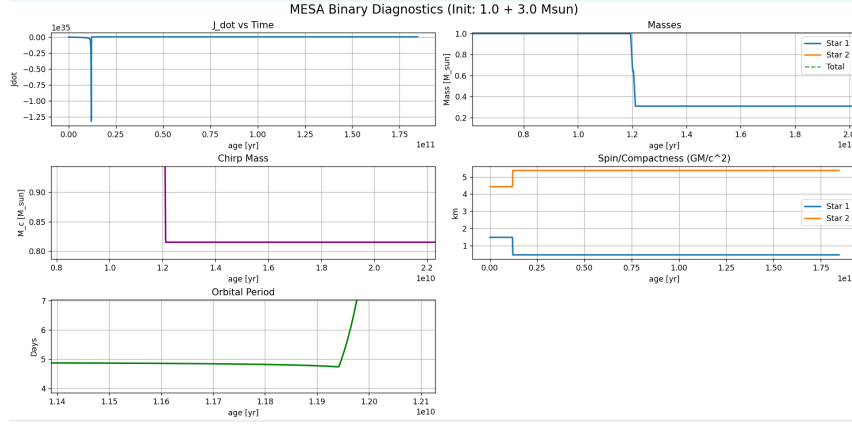


Figure 3: Parameter evolution of a binary system with $M_1=1 M_\odot$ and $M_2=2 M_\odot$ using MESA, focusing on period shows a decrease in period followed by a sharp increase.

This is the same plot for the same above binary system with period zoomed in to see the change in period, the period decreases very sharply between 1.19 to 1.20 billion years - in even smaller timesteps the period would show an even sharper decrease. This could imply that initially the period was 5 days which decrease to the point where both stars touched and start to accrete matter, which triggers a runaway flow of matter between the stars which could create a kick which increases the separation between the stars, hence the increase in period we see which lasts until the system becomes stable again and from then on the stars are not massive enough to further any instability flag in them and continue to evolve without any interaction between them.

Our goal however, was to create a system with NS and a WD in orbit with each other, this would require one of the stars to have a mass between the ranges $(1.4 - 8) M_\odot$ which is roughly the theoretical limit of the progenitor of a WD and the other star should form a NS the progenitor of which has roughly the mass range between $(8-15) M_\odot$. This was shown to become quite a difficult task, as the evolution of stars upto NS-WD system is not a trivial task. Initially, while evolving the stars i was under two misconceptions about MESA

- MESA evolves both stars and it's structure even with the `evolve_both_stars` flag given to be false, hence i was running the binaries with this flag turned false. The reason for this was when i turned this flag to be true, the simulation would end a lot earlier, hence i was under the wrong assumption that MESA was unable to handle both their evolution because of mass discrepancies and not that if this flag was kept true, both stars were treated as a self-consistent system which evolved and

changed its structure and radius with its age, hence triggering very complex phases like Roche-lobe overflow (RLO) and SNe events which i'll talk about later. This is changed in the planned future runs using cluster to speed up the process and using the `binary_c` code which would help us concretize the results from MESA or tell us if using MESA is even necessary for a cosmologically correct population according to the IMF of the cluster/ Universe.

- I was under the incorrect assumption that MESA handles evolution of these complex binaries all the way through from initial masses to NE_binaries and all the way to their mergers, MESA infact cannot do that, it cannot handle the immediate after effect of a SNe event after which a NS is born.

This was done after we had done a couple 20-30 runs on cluster and gotten data for them, after plotting them using the python pipeline we created. We chose masses between the above specified ranges for all of these runs keeping the `evolve_both_stars` flag to be false. This was done in order for evolution of binary systems to reach the current age of the universe - 13.6 billion years, but what was observed was that the simulation ended long before we reached that age.

5 Multiple Binary Runs Using Cluster

The results from some of such multiple runs are shown below:

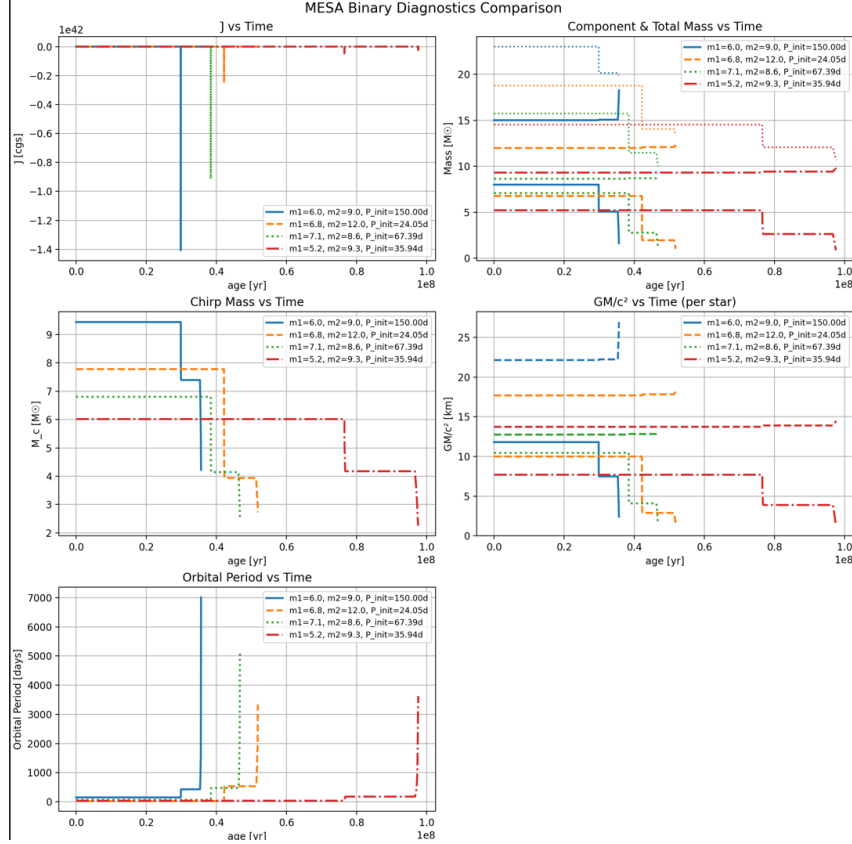


Figure 4: Parameter evolution of multiple binary systems overlaid on top of each other to show their evolution with time

All of these systems show a similar behavior to the first system we evolved, which clearly shows something is wrong with the way we are evolving the systems, the problem as mentioned above was thinking MESA evolved the system all the way through to NS-WD state and further upto it's merger which was incorrect assumption. The correct way to evolve stars in MESA to produce such systems is: First evolve the stars keeping the `evolve_both_stars` flag to be `.true`. This is done upto the point where one of the stars, the more massive one undergoes a SNe. The profile of this star and its companion is saved for future runs. The data we get from the SNe is then used to infer the "kick" received by the companion star to model the system as a still bound or a detached binary. This system is then further evolved after the SNe and the `evolve_both_stars` flag is turned `.false`, this means the NS is now treated as a point source and all the processes like mass-transfer, ROL, angular mometum transfer are considered - MESA does that for us while evolving

the stars- evolving the NS + a massive star as an X-ray binary up until the point where we reach the common envelope (CE) phase or if the stars are sufficiently far apart, upto the age where the other star's reaches a WD phase, this is the system we are looking for, that is a NS-WD system, the system of our study. Due to time constraints and the time it takes for MESA to simulate, this will be done in the very near future.

6 Conclusion and future direction

Multiple binary systems were evolved using MESA in the PARAM_RUDRA cluster of IIT Bombay with different initial masses and periods which could produce NS-WD binary systems. Initially it was assumed MESA takes care of the entire evolution for the system by evolving both the stars into the NS-WD phase and to later merger, but this was found out to be incorrect from the data produced, as we can see from the plots. The system reaches at most 10-15 million years after which MESA terminates and the evolution stops. This was checked with multiple periods with multiple masses and was found to be consistent with each other. In the very near future we plan to use the method described as, evolve both stars upto the SNe kick and determine the model of the kick from the data MESA provides, then turn the heavier star that triggered the SNe as a point by turning the `evolve_both_stars` flag to be false, and evolve the system into the X-ray binary and CE phase, after which confirmation of a simulated NS-WD system can be concretely made. Using `binary_c` to simulate such systems and produce a large population, setting up `binary_c` and checking the method needed to create a NS-WD system is of key importance (1). After creating a single NS-WD system in MESA and `binary_c` we can compare the physical processes and timescale of formation of these systems. Creating an entire population for NS-WD through both these methods, we can do comparative studies for these codes creating comprehensive results and creating a path for future studies of these non-trivial systems

7 Second Term of Project work

This is the continuation of the work done previously. All simulations of evolution are done using `binary_c` (13)

7.1 `binary_c`

`binary_c` is a rapid stellar evolution code that can be used to evolve multiple binaries very efficiently. It involves all the evolution mechanism of single star evolution with binary evolution prescriptions like mass transfer, mass accretion, and common envelope (CV) (13). what is even more important for us is that it follows a prescription that determines the stages of star during it's evolution from ZAMS, when the simulation ends, we can be make sure what type of stars or stellar remnants we have, therefore determining if NS-WD binaries have formed or not.

The prescription or code that the code follows is as below:

Code	Evolutionary Phase
0	MS star $M \lesssim 0.7$ (deeply or fully convective)
1	MS star $M \gtrsim 0.7$
2	Hertzsprung Gap (HG)
3	First Giant Branch (GB)
4	Core Helium Burning (CHeB)
5	Early Asymptotic Giant Branch (EAGB)
6	Thermally Pulsing AGB (TPAGB)
7	Naked Helium Star MS (HeMS)
8	Naked Helium Star Hertzsprung Gap (HeHG)
9	Naked Helium Star Giant Branch (HeGB)
10	Helium White Dwarf (HeWD)
11	Carbon/Oxygen White Dwarf (COWD)
12	Oxygen/Neon White Dwarf (ONeWD)
13	Neutron Star (NS)
14	Black Hole (BH)
15	Massless remnant

Table 2: Stellar evolutionary phases and their corresponding codes (13).

The setup of the problem is similar to what we have done previously; we first take sets of initial masses initially chosen to confirm the hypothesis that they will form a WD or an NS. The mass ranges for Primary M1 is [0.8 to 8], which at the end of its evolution should form a WD and the mass range for secondary M2 is [8 to 15] which at the end of its evolution should form an NS.

8 Downloading, Installing and Running binary__c

Initially, The setup is done a local pc. We will be running 10^6 systems using IIT's supercomputer ParamRudra. To install binary__c we follow the instructions provided in the binary__c documentation ([binary__c](#)).

Before Installing binary__c we first create the environment and dependencies that would be required for it to run, We follow the instructions below to setup the environment:

Setting binary__c directory: Add the following to your .bashrc or equivalent shell profile:

```
export BINARY_C=$HOME/progs/stars/binary__c
```

and **restart the shell** so that the environment variable BINARY_C is defined.

Creating source file for binary__c to manually control environment of shell: We

create a .sh file named LIBRARY_PATH.sh, sourcing this will setup the binary__c environment in the shell. Set up LIBRARY_PATH, LD_LIBRARY_PATH, and PATH in LIBRARY_PATH. For example, add the following to LIBRARY_PATH if you are using bash:

```
export LD_LIBRARY_PATH=$HOME/lib:$BINARY_C/src
export LIBRARY_PATH=$HOME/lib
export PATH=$PATH:$HOME/.local/bin:$HOME/bin
```

and **restart the shell**.

Probably required: Update your meson and ninja:

```
pip3 install --upgrade meson
pip3 install --upgrade ninja
```

Probably required: If you can, install librinterpolate. binary__c includes a version of librinterpolate as a backup, but it's useful to build and install it so you get the latest version.

```
mkdir $HOME/lib $HOME/git
cd $HOME/git
git clone https://gitlab.com/rob.izzard/librinterpolate.git
cd $HOME/git/librinterpolate
meson setup --prefix=$HOME --buildtype=release builddir
ninja -C builddir install
```

We will be using `binary_c-python` to run the simulation for our population studies, so installing `librinterpolate` is required.

Run the following command to download `binary_c`:

```
git clone https://gitlab.eps.surrey.ac.uk/ri0005/binary_c.git
```

Once this is finished, enter the new directory:

```
cd binary_c
```

We then need to configure the build tree. We do this using `meson`:

```
meson setup builddir
```

This creates a directory called `builddir` in which the compiler is set up. Next, go into the `builddir`:

```
cd builddir
```

And finally, build `binary_c`:

```
ninja
```

If this succeeds, you should have a binary called `binary_c` in your `builddir`. Test it with:

```
./binary_c
```

It should automatically run and check for systems and give logs as it is building and testing itself; for me, the build and test was completed with the output below.

```
{
Warning: No parse function set. Make sure you intended to do this.
Write grid code to /tmp/binary_c_grid_765663f8dd5c4f779bde0f9d663b9719.py [dry_run = True]
Doing a dry run of the grid.
Grid has handled 738 stars with a total probability of 1.0077

*****
*           Dry run           *
*   Total starcount is 738    *
*   Total probability is 1.0077 *
*****

setting up the system_queue_filler now
Write grid code to /tmp/binary_c_grid_765663f8dd5c4f779bde0f9d663b9719.py [dry_run = False]
Signalling processes to stop

263/738  35.6% complete 11:03:22 ETA=    9.1s tpr=1.91e-02 ETF=11:03:31 mem:
176.7MB M1=1.2 M2=0.38 P=22

329/738  44.6% complete 11:03:27 ETA=   31.0s tpr=7.58e-02 ETF=11:03:58 mem:
118.8MB M1=1.2 M2=0.75 P=3.6e+7
```

Figure 5: Build and test for binary_c

After it has done build and test, you should see an output similar to one below:

```
*****
*               Process 0 finished:               *
* generator started at 2026-01-21T11:03:19.728766 *
* generator finished at 2026-01-21T11:04:06.460532 *
*               total: 46.73s                      *
*               of which 46.31s with binary_c      *
*               Ran 738 systems                   *
*               with a total probability of 1.0077  *
*               This thread had 0 failing systems  *
*               with a total failed probability of 0 *
* Skipped a total of 72 zero-probability systems  *
*               *                                  *
*****

*****
* Population-765663f8dd5c4f779bde0f9d663b9719 finished! *
*               The total probability is 1.0077.          *
* It took a total of 47.10s to run 738 systems on 1 cores *
*               = 47.10s of CPU time.                   *
*               Maximum memory use 176.691 MB            *
*****

No failed systems were found in this run.
Normalize to mass_in_stars = 0.7920163642110047 Msun
outfile= /tmp/ensemble_output.json.bz2
ensemble has been written to /tmp/ensemble_output.json.bz2
binary_c install script finished
```

Figure 6: Testing completed message: This is for a laptop which has a 4 core intel i7 cpu, only one is being utilized in building and testing binary_c

8.0.1 Testing with (test binary stellar evolution) ./tbse

After finishing installation and building binary_c, we use ./tbse to check and evolve a single binary system and observe the output.

For me, the location of tbse file is:

```
/home/ingenuity/progs/stars/binary_c
```

We change the mass and period parameters in the .tbse file to test for a single evolution, we choose $M1=15 M_{\odot}$ and $M2=7 M_{\odot}$ with a period of 1000 days and let the system evolve, to see the output we run ./tbse with the following line

```
./tbse --log_filename /dev/stdout
```

The thing of importance is we let the period relax, from initially taking it to be 50 days, during this evolution no NS-WD system was formed, and the end product was just

a stellar remnant. We let the system evolve with a period of 1000 days, let the system run for 14 billion years to capture the evolution within the lifetime of the universe. and confirm the formation of a NS-WD binary within 14 billion years.

The output from the ./tbse run with $M_1=15 M_\odot$, $M_2=7 M_\odot$ and period=1000 days is as below:

9 Running a population with binary_c-python

binary_c-python is used for creating and evolving a large number of initial sets of masses and periods from ZAMS to end of their evolution, it is exclusively used from this point on to create populations of NS-WD binaries from initial sets of parameters.

9.1 First "simpler" population run

We use binaryc-python to run the systems with following initial parameters: $M_1=[0.2 \text{ to } 8] M_\odot$ $M_2=[8 \text{ to } 15] M_\odot$ with a gap of 0.2, so the pairing of each sets of masses is like this: [0.2, 8], [0.4, 8] ..., [0.2, 9], [0.4, 9] ... A really **wrong** assumption taken here is that the period is same for all of these systems, I chose 2000 days as the initial period for all of these systems, which ofcourse is a "bold" assumption, but i chose it to let all of the systems evolve for as long as possible before triggering a merger event with a supernova, To let the star evolve as it is orbiting the secondary and go through all possible cycles, so we get the expected final evolved stars in an orbit. The expectation was that the stars with the mass range from [0.2 to 8] $M_\odot$ would form a WD and the stars with the mass range from [8 to 15] $M_\odot$ would form NS.

The results of all of the runs were collected in a folder, all the runs were formatted to give columns as output, with Masses, periods, eccentricity and type of the stars at each timestep. The data was analyzed using a python pipeline to detect the formation of NS-WD binaries from this run.

The result of analyzing the data is as follows:

```

Scanning data files for NS-WD systems...
Scanned 0/296 files...
Scanned 29/296 files...
Scanned 58/296 files...
Scanned 87/296 files...
/home/ingenuity/progs/stars/binary_c/NS-WD_finder.py:38: DtypeWarning: Columns (0) have mixed types. Specify dtype option on import or set low_memory=False.
  df = pd.read_csv(filepath, sep=r'\s+')
Scanned 116/296 files...
Scanned 145/296 files...
Scanned 174/296 files...
Scanned 203/296 files...
Scanned 232/296 files...
Scanned 261/296 files...
Scanned 290/296 files...
-----
Scan Complete.
Found 217 NS-WD systems.
Summary saved to: nswd_summary.csv
-----
      File  Init_M1  Init_M2  Init_Period  ...  Final_Ecc  Final_K1  Final_K2  Final_Sep
214  8_1.4_2000.dat    8.0    1.4    2000.0  ...  3.901840e-09    13    11  1114.7300
83   8_1.6_2000.dat    8.0    1.6    2000.0  ...  0.000000e+00    13    11   937.4350
109  8_1.8_2000.dat    8.0    1.8    2000.0  ...  9.999810e-06    13    11    40.5593
185  8_2.0_2000.dat    8.0    2.0    2000.0  ...  9.998510e-06    13    11    24.4566
208  8_2.2_2000.dat    8.0    2.2    2000.0  ...  9.992530e-06    13    11    16.3484

```

Figure 7: Output of first runs with masses paired together in mass range $[0.2 \text{ to } 15] M_{\odot}$ with constant period of 2000 days

We did a total of 296 runs with these parameters and saw that 217 NS-WD binary systems formed, which confirms our hypothesis that the mass ranges chosen are correct for the formation of these binaries.

What this "naive" way of doing things does not capture is the actual statistics of star formation that is the correct mass and period distribution. Hence, we work to make our model more correct by feeding the correct parameters derived from already well established theories and observations.

10 The final setup of runs and results

Initially, we will be using a local pc environment, and for the final runs with a population of greater than millions, we will be using IITB's supercomputer, ParamRudra.

10.1 10,000 population setup

We will use the following Physics to choose our initial parameters, M1, M2 and Period.

M1 -> The primary mass is taken from a distribution of masses according to Kroupa IMF(8), as mentioned above it is a two-piece broken power law given as:

$$\begin{aligned}\alpha_1 &= 1.3 \pm 0.3, & , & \quad 0.08 \lesssim m/M_\odot \lesssim 0.5, \\ \alpha_2 &= 2.3 \pm 0.5, & , & \quad 0.5 \lesssim m/M_\odot \lesssim 150.\end{aligned}\tag{1}$$

This distribution is used to sample the masses by using the python pipeline as below:

```
1 def vectorized_kroupa(n_required, m_min=0.08, m_max=150.0):
2     """
3     Highly optimized, vectorized Kroupa IMF sampler.
4     Enforces absolute physical stellar mass limits (0.08 to 150 M_sun).
5     """
6     # Safety Clamp: Enforce the physical limits of the universe
7     m_min = max(0.08, m_min)
8     m_max = min(150.0, m_max)
9
10    accepted = []
11
12    # Max probability occurs at the minimum mass
13    max_prob = m_min**(-1.3) if m_min < 0.5 else 0.5 * m_min**(-2.3)
14
15    while len(accepted) < n_required:
16        # Generate a large batch to account for rejections
17        # Since 150 M_sun stars are incredibly rare, we need a larger batch
18        # multiplier
19        batch_size = int((n_required - len(accepted)) * 10)
20
21        m_guess = np.random.uniform(m_min, m_max, batch_size)
```

```

21     p_guess = np.random.uniform(0, max_prob, batch_size)
22
23     # Vectorized calculation of the Kroupa PDF
24     pdf = np.where(m_guess < 0.5, m_guess**(-1.3), 0.5 * m_guess**(-2.3))
25
26     # Boolean mask to keep only accepted masses
27     mask = p_guess < pdf
28     accepted.extend(m_guess[mask])
29
30     # Return exactly the number requested
31     return np.array(accepted[:n_required])

```

The basic logic of the code is that it creates a sampler for masses between 0.08 to $15 M_{\odot}$ using the kroupa IMF: The work flow of the code is as follows:

1. **Vectorized Random Generation:** We use a vectorized model to sample the masses; it uses vectorization. A large batch of random candidate masses (m_{guess}) and candidate probabilities (p_{guess}) are generated simultaneously using uniform distributions:

$$m_{\text{guess}} \sim \mathcal{U}(m_{\min}, m_{\max})$$

$$p_{\text{guess}} \sim \mathcal{U}(0, p_{\max})$$

2. **The Rejection Filter (Masking):** For the entire array of candidate masses, the exact curve value $\xi(m_{\text{guess}})$ is computed simultaneously. Each candidate is then evaluated against its corresponding uniform probability coordinate:

- **Acceptance:** If $p_{\text{guess}} < \xi(m_{\text{guess}})$, the candidate falls within the target probability distribution and is appended to the accepted population array.
- **Rejection:** If $p_{\text{guess}} \geq \xi(m_{\text{guess}})$, the candidate represents a non-physical probability and is discarded from memory.

3. **Iterative Replenishment:** Steps 2 and 3 are repeated dynamically. The algorithm calculates the deficit between the current number of accepted masses and the

total requested population (N_{req}), generating subsequent vectorized batches until the quota is exactly satisfied.

Plotting the PDF and masses as a log log plot gives the following result:

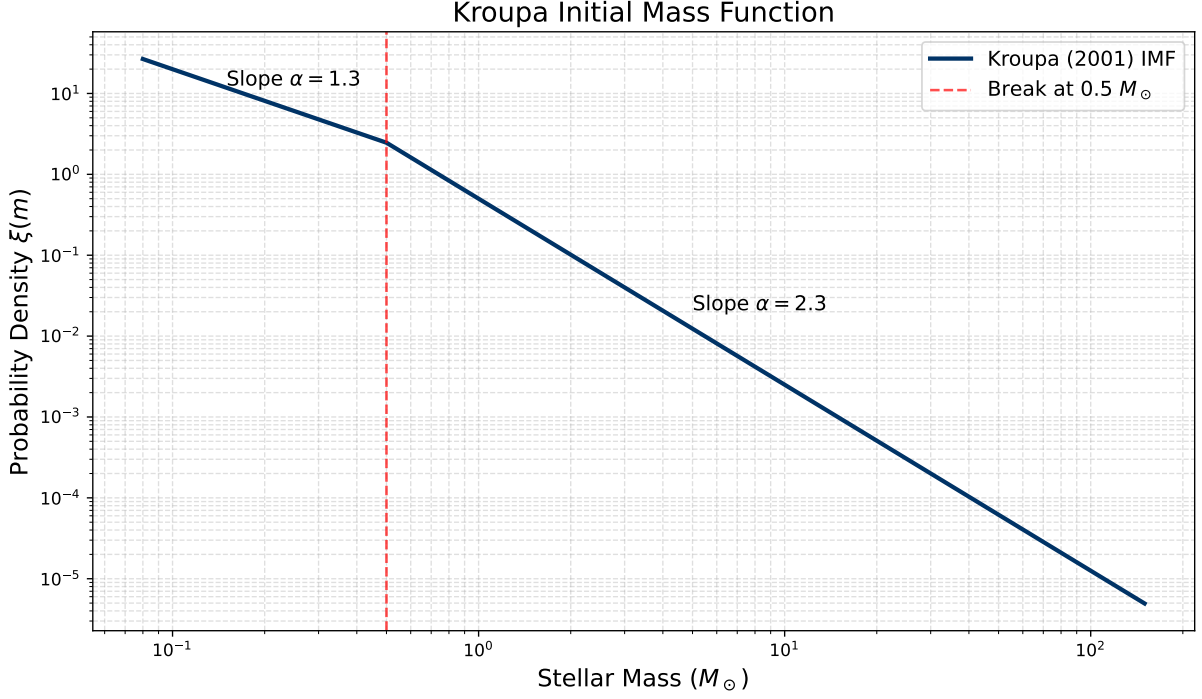


Figure 8: log log plot of Kroupa IMF based on our rejection sampling

This was the most complex part of the sampling, a mass ratio $M2/M1 = q$ is sampled from $[0.1 \text{ to } 1]$ and $M2$ is calculated using $q * M1$, This is sampled uniformly since q is picked uniformly between 0.1 to 1 but the sanctity of the kroupa IMF is shown in $M2$ as well as $M1$ is sampled from Kroupa IMF.

For periods we chose a simpler model, we randomly and uniformly sample periods from $10 * (R1+R2)$ to $100 * (R1+R2)$, where the Radii are calculated based on the ZAMS masses from the prescriptions based on Tout et al (1996) (15) which follow the power law below:

$$R \approx \begin{cases} M^{0.80} & \text{if } M < 1.0 M_{\odot} \\ M^{0.57} & \text{if } M \geq 1.0 M_{\odot} \end{cases} \quad (2)$$

With this pipeline, we create 10,000 binaries and save the initial data we have generated as a .csv file, that we can feed to binary_c and run all the initial parameters in binary_c.

After engaging the run we let `binary_c` evolve the systems and save the output file as a `.dat` file in a different folder so it can be accessed easily later to analyze the results of evolution.

The output file, a single `.dat` file has columns with masses, periods and types of stars, the types of stars can be coded as a number later to convert into hdf5 file for better acces. we can run a similar pipeline here and identify the combination of masses and periods that have successfully formed a NS-WD binary.

The result of running the pipeline after getting the data from `binary_c` is shown in the figure below:

```

--- Processed 7000/10000 systems... ---
SUCCESS: NS-WD formed! Source File -> 13.984_10.078_1.09028.dat
SUCCESS: NS-WD formed! Source File -> 14.11_8.978_0.340964.dat
SUCCESS: NS-WD formed! Source File -> 14.148_7.813_1.27029.dat
SUCCESS: NS-WD formed! Source File -> 14.621_9.202_0.0656901.dat
--- Processed 8000/10000 systems... ---
--- Processed 9000/10000 systems... ---
SUCCESS: NS-WD formed! Source File -> 7.088_6.866_0.0771359.dat
--- Processed 10000/10000 systems... ---

=====
SCAN COMPLETE
=====

Total Systems Analyzed: 10000
Total NS-WD Binaries Formed: 5
Formation Efficiency: 0.05%

```

Figure 9: Result of 10,000 runs with sampling of masses done from Kroupa IMF and uniform periods sampled from $10 * (R1+R2)$ to $100 * (R1+R2)$

From the results, we can see that out of the 10,000 systems, only 5 NS-WD binaries have formed, which is a much better statistical number. The time it took to run all these systems in my machine which has an intel i7 9th gen processor with 4 cores was around 1.5 hours, which is a very good number considering the number of systems I was evolving and the computer that was used for it.

The plot of 4 of these binaries is given below:

Neutron Star - White Dwarf Progenitor Evolution

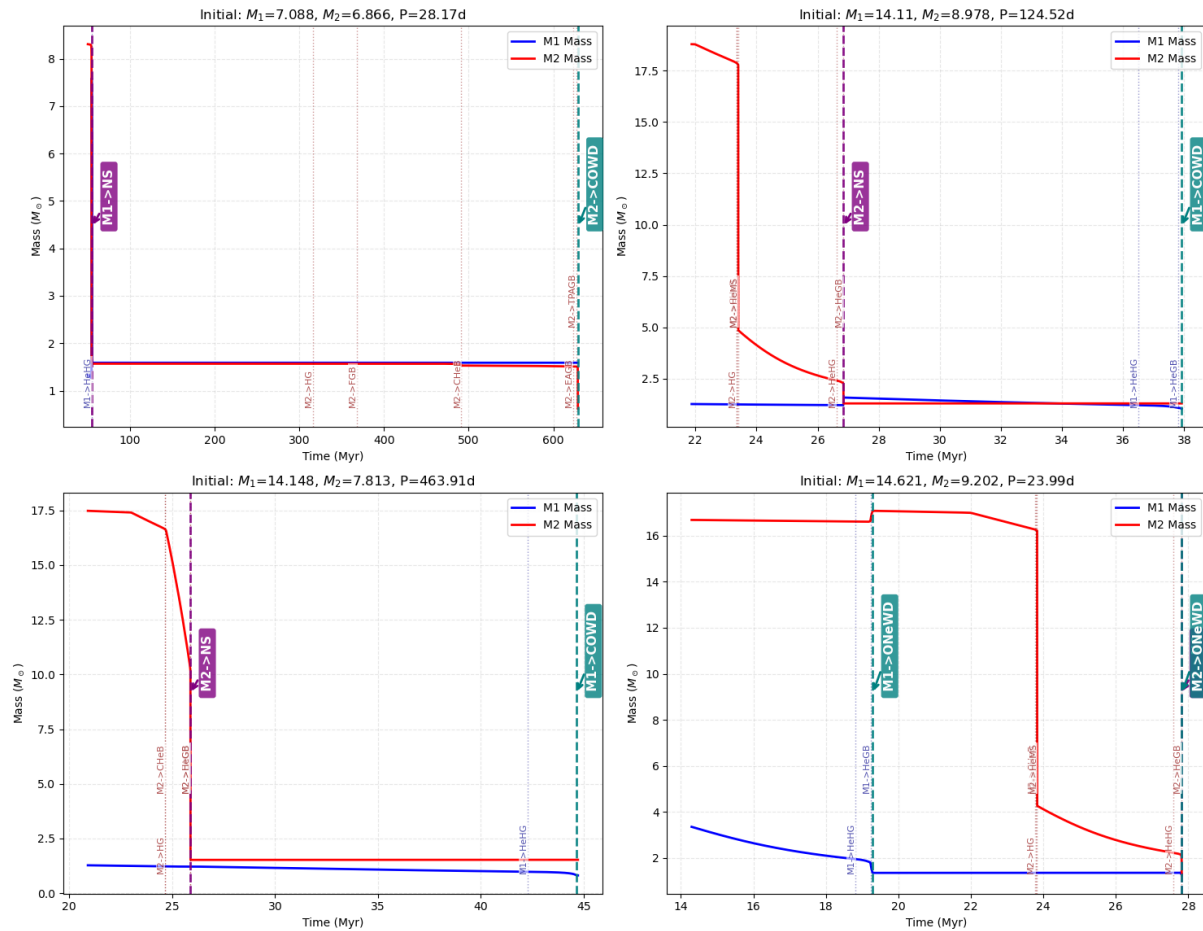


Figure 10: Plot of evolution of masses of 4 systems which formed NS-WD binaries; the points where there was a type change is also present in the plots

11 Run with 10 million binaries

The next step in our research is to scale the simulation to a larger population of initial parameters and run these said parameters through `binary_c` and see their evolution, this will give a much better dataset to work with to figure out the merger rates of these systems which is the big goal of this thesis.

To do this we follow some sets of crucial steps on IITB’s cluster, ParamRudra which are similar to but very different from doing it on my computer, as we don’t have admin access in Param, we use some workarounds and some modules like spack which are present in Param to setup, install and run `binary_c`. The exact steps will be given in the appendix of this thesis.

11.1 The setup and planned runs

The setup of the problem is exactly similar to one where we used `binary_c` to create initial population of binaries and ran them using `binary_c-python`, the difference in cluster is that I am using `binary_c` with a modified `logging_every_timestep` function. The exact function I am using will give me columns that contain, masses, periods, types, eccentricities, radii and Roche Lobe Overflow (RLOF) for each of the run and save all the different runs as a `.dat` file following the format `[M1_M2_period]`. The idea to add Radii and RLOF as columns was to check during evolution when the star fills the Roche lobe and stars inspiral, this could in practice give the time at which the objects merge and also the time when they would start to produce GW during their inspiral.

The exact `logging_every_timestep` which is contained in the exact path home:

`/IITB/CompnalAstrphyNRelvity/24n0294/progs/stars/binary\_c/src/logging_every_timestep`

The exact `logging_every_timestep` code is given below:

```
#undef HRDHEADER
#define HRDHEADER "HRDdiff"
    Printf(HRDHEADER " %g %g %d %g %g %d\n",
        stardata->common.zero_age.mass[0],
        stardata->model.time,
        stardata->star[0].stellar_type,
        HRdiagram(0),
        updated
    );
}

if(timestep_fixed_trigger(stardata, FIXED_TIMESTEP_ARG))
{
    Printf("Log at chosen time %g\n", stardata->model.time);
}

/* Master Thesis Custom BSE Output */
if (stardata->model.time >= 0.0) {
    printf("BSE: %g %g %g %d %d %g %g %g %g %g %g\n",
        stardata->model.time,
        stardata->star[0].mass,
        stardata->star[1].mass,
        stardata->star[0].stellar_type,
        stardata->star[1].stellar_type,
        stardata->common.orbit.period * 365.25,
        stardata->common.orbit.eccentricity,
        stardata->common.orbit.angular_momentum,
        stardata->star[0].radius,
        stardata->star[1].radius,
        stardata->star[0].roche_radius,
        stardata->star[1].roche_radius);
}
}
```

Figure 11: The custom handles to print the columns for masses, periods, type, eccentricity, radii and RLOF

We run the MC initial population pipeline on the cluster to create a .csv file that has 10 million initial parameters of M1, M2 and periods. We then create a custom Python pipeline that feeds these custom initial parameters into `binary_c` and evolves all the systems, saving outputs as described above.

The job is submitted with the following parameters on the cluster, The process ran with 48 cores:

```
#!/bin/bash
#SBATCH --job-name=binary_c_grid
#SBATCH --output=logs/grid_%j.out
#SBATCH --error=logs/grid_%j.err
#SBATCH --nodes=1
#SBATCH --ntasks=1
#SBATCH --cpus-per-task=8
#SBATCH --mem=16G
#SBATCH --partition=small
```

Figure 12: The parameters that were used for the simulation, a total of 48 cores were used in the simulation.

with the current progress, we were able to evolve about 1.5 million systems in 13 hours.

Hurley et al (2002) (13) have taken a constant star formation rate with the assumptions, that one binary with $M_1 \geq 0.8M_\odot$ is born in the galaxy per year, giving:

$$S \int_{0.8}^{\infty} \Phi(m) dm = 1, \quad (3)$$

which gives $S = 7.6085 \text{ yr}^{-1}$. (13). They argue that this is in rough agreement with the birth rate of WDs in the Galaxy which is

$$\chi_{WD} \approx 2 \times 10^{-12} \text{ pc}^{-3} \text{ yr}^{-1}$$

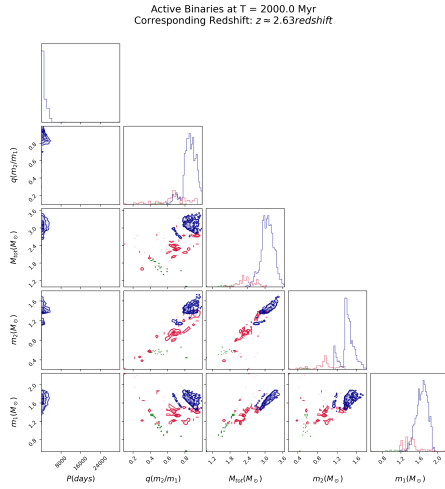
They make the assumption that only stars with $M \geq 0.8M_\odot$ can form a WD and all stars that are formed are in binaries.

There are different models which they have created based on different prescriptions of common envelope efficiency parameter Tides being on or off, metallicity set to $z=0.02$, for our current simulation we are using Model A.

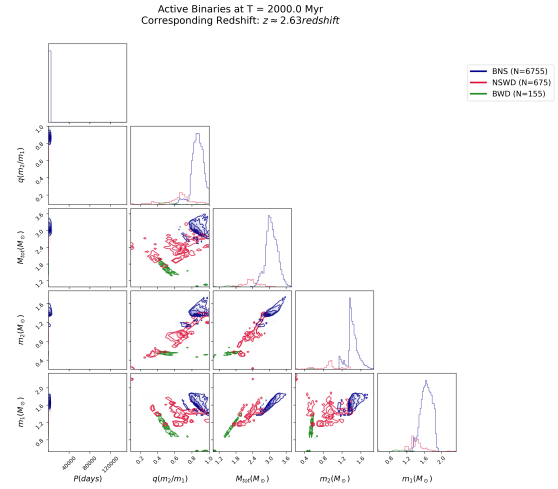
12 Results with 1 million binaries

12.1 Corner plots of population

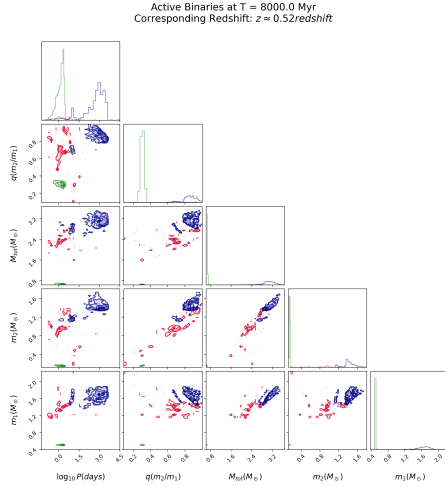
Corner plots are essential in understanding how the different parameters are distributed and correlated in our population, and it gives us control to check and change the parameters accordingly. Here we show Corner plots at different times - 1 billion years to 14 billion years, we can clearly see how the individual masses, periods and mass ratios are distributed.



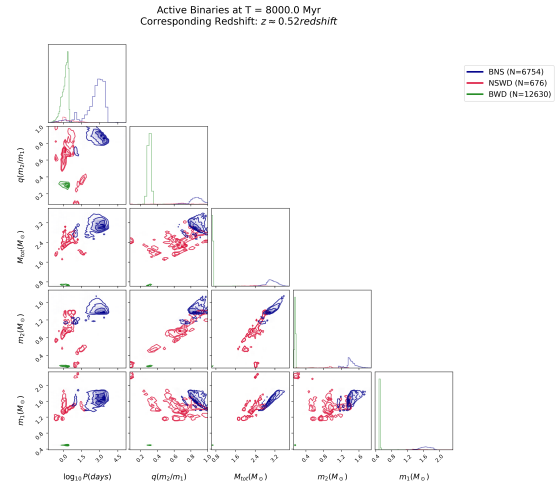
(a) Corner plot of 1 million simulated binary systems at 2 billion years



(b) Corner plot of 6 million simulated binary systems at 2 billion years



(c) Corner plot of 1 million simulated binaries at 6 billion years



(d) Corner plot of 6 million simulated binaries at 6 billion years

12.2 Plot showing Population of binaries as a function of time

The plot showing the number of different binaries with respect to time is shown below:

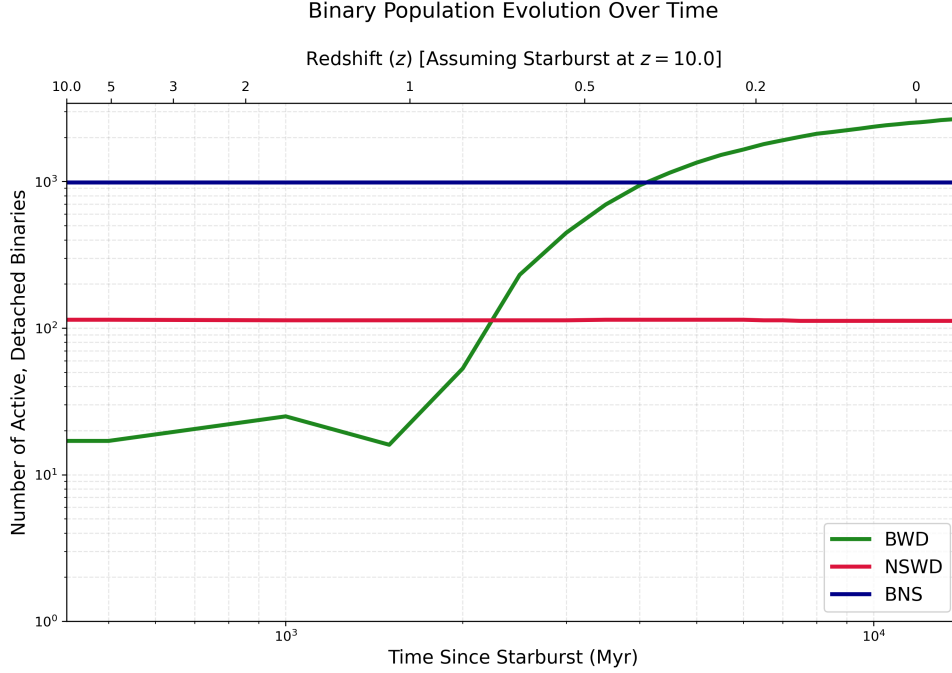


Figure 14: Number of binaries as a function of time (t) and red shift (z)

13 Final conversion and merger rate calculation

We transform the merger rate per simulated system to merger rate $\text{Gpc}^{-3}\text{yr}^{-1}$, In order to do this we follow the methodology described by Dominik et al (2013) (16). This methodology convolves the delay-time distribution of the simulated binaries with an empirical cosmic Star Formation Rate (SFR) history, which we are taking from Wilkins et al. (2019) (17), yielding the rest-frame merger rate density $\mathcal{R}$ in units of $\text{Gpc}^{-3}\text{yr}^{-1}$. We compare the merger rate density of our simulated BNS systems and compare it with LIGO bounds for BNS merger rates from the LVK collaboration (18)

13.1 Simulated Mass Normalization

Population synthesis codes generate a discrete number of Zero Age Main Sequence (ZAMS) binary systems, N_{total} . We need to convert this merger rate per simulated binary denoted as M_{sim} , to merger rate $\text{Gpc}^{-3} \text{yr}^{-1}$

We have Kroupa IMF for generating our initial masses, the average mass of a generated system $\langle M_{\text{sys}} \rangle$ is calculated by integrating the IMF over the simulated mass, given as

$$M_{\text{avg}} = \frac{\int_{M_{\text{min}}}^{M_{\text{max}}} M \xi(M) dM}{\int_{M_{\text{min}}}^{M_{\text{max}}} \xi(M) dM}$$

where $\xi(M)$ is the Kroupa IMF. For a Kroupa IMF with $M_{\text{min}} = 0.2 M_{\odot}$ and $M_{\text{max}} = 15 M_{\odot}$.

The total simulated ZAMS mass is therefore:

$$M_{\text{sim}} = N_{\text{total}} \times \langle M_{\text{sys}} \rangle \quad (4)$$

Dividing the number of successful mergers by M_{sim} yields the formation efficiency: the number of merging compact binaries produced per solar mass ($M_{\odot}$) of star-forming gas.

13.2 Converting the Continuous integral to the Discrete Cosmological Summation

To find merger rates $\text{Gpc}^{-3}\text{yr}^{-1}$ we use the integration:

$$\mathcal{R} = \int_0^{t_{\text{Hubble}}} \text{SFR}(13.8 - t_d) \left(\frac{1}{N_{\text{total}} \cdot M_{\text{avg}}} \frac{dN_{\text{merge}}}{dt_d} \right) dt_d$$

We need to convert this integral to a discrete sum to get the final merger rates; for this we utilize the exact delay time, $t_{\text{delay},i}$, of each successfully merging binary i . The delay time is defined as the total interval from ZAMS to final coalescence.

To calculate the local merger rate density today (at redshift $z = 0$), we evaluate the universe at its current age, $t_{\text{Hubble}} \approx 13.8$ Gyr. A binary observed merging locally today must have been formed at a lookback time equal to its specific delay time. Thus, the exact cosmic birth timestamp for the i -th binary is given by $(t_{\text{Hubble}} - t_{\text{delay},i})$.

At this birth timestamp, the available volume density of star-forming gas is dictated by the cosmic star formation rate history, $\text{SFR}(t)$. By evaluating the empirical SFR at the exact birth time of each simulated binary, we scale the individual merger event by the cosmological star formation history. We use the curve of SFR from Wilkins et al. (2019) (17). To get the data from the curve between

X-

$$\log_{10}(1+z)$$

and Y-

$$\log_{10}\left(\frac{\text{SFRD}}{M_{\odot} \text{ yr}^{-1} \text{ Mpc}^{-3}}\right)$$

we use webplotdigitizer to get the data from their curve and convert the units to

X-

$$\text{Cosmic Age (Gyr since Big Bang)}$$

and

Y-

$$\text{SFR Density } (M_{\odot} \text{ yr}^{-1} \text{ Mpc}^{-3})$$

We use this curve for SFR at each different timestep

Following the discrete rest-frame rate density equations defined by Dominik et al. (2013) (16), the total local volumetric merger rate density $\mathcal{R}_{z=0}$ is calculated as the sum of these individual cosmological weights, normalized by the total simulated mass:

$$\mathcal{R}_{z=0} = \frac{1}{M_{\text{sim}}} \sum_{i=1}^{N_{\text{merge}}} \text{SFR}(t_{\text{Hubble}} - t_{\text{delay},i}) \quad (5)$$

13.3 Dimensional Scaling

Empirical Star Formation Rate histories are typically reported in units of $M_{\odot} \text{ yr}^{-1} \text{ Mpc}^{-3}$. Because M_{sim} is expressed in units of $M_{\odot}$, the mass dimensions in Equation 5 naturally cancel out:

$$\frac{M_{\odot} \text{ yr}^{-1} \text{ Mpc}^{-3}}{M_{\odot}} = \text{yr}^{-1} \text{ Mpc}^{-3} \quad (6)$$

To align our theoretical predictions with the metrics used by the LIGO-Virgo-KAGRA (LVK) collaboration, the resulting density is scaled from cubic megaparsecs to cubic gigaparsecs. Since $1 \text{ Gpc} = 10^3 \text{ Mpc}$, the volume conversion factor is:

$$1 \text{ Mpc}^{-3} = 10^9 \text{ Gpc}^{-3} \quad (7)$$

Multiplying the result of Equation 5 by 10^9 yields the final, observable local merger rate density in the canonical units of $\text{Gpc}^{-3} \text{ yr}^{-1}$.

14 Conclusion and Results

15 Conclusion

We find the merger rates for different binaries as part of our conclusion to this study, the result is as follows:

```
-> Total simulated ZAMS binaries: 998,714
-> Total simulated mass (M_sim): 1,194,461.94 M_sun
```

Final Observable Merger Rates ($z = 0$)

Population	Valid Mergers (n)	Rate Density ($\text{Gpc}^{-3} \text{ yr}^{-1}$)
BNS	7	79.29
NSWD	10	166.42
BWD	191	6,253.47

The simulated Binary Neutron Star (BNS) merger rate density in the local universe ($z = 0$) was found to be $\mathcal{R}_{\text{BNS}} \approx 79.29 \text{ Gpc}^{-3} \text{ yr}^{-1}$. This is within the limits of $10 - 1700 \text{ Gpc}^{-3} \text{ yr}^{-1}$ initially derived from the LVK GWTC-3 catalog (18). Furthermore, this theoretical prediction aligns perfectly with the newly updated, tighter constraints of $7.6 - 250 \text{ Gpc}^{-3} \text{ yr}^{-1}$ established by the GWTC-4.0 catalog(18)

15.0.1 Final Results

We simulated 1 million binaries and evolved them with `binary_c`, We checked for the merger rates for different binaries. The BNS merger rate we got from our simulation is $\mathcal{R}_{\text{BNS}} \approx 79.29 \text{ Gpc}^{-3} \text{ yr}^{-1}$, which when compared with the merger rate bound from LVK collaboration which is $7.6 - 250 \text{ Gpc}^{-3} \text{ yr}^{-1}$. We find the corresponding merger rate for NSWD binaries to be $166.42 \text{ Gpc}^{-3} \text{ yr}^{-1}$

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